

ETIOPATHOGENESIS OF FELINE IDIOPATHIC CYSTITIS*

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COMORBID CONDITIONS IN CATS WITH
FELINE IDIOPATHIC CYSTITIS

PATHOPHYSIOLOGY OF FELINE
IDIOPATHIC CYSTITIS

TREATMENT STRATEGIES
CONCLUSION

Feline idiopathic cystitis (FIC) has been characterized as a chronic lower urinary tract syndrome of unknown cause and with no widely accepted treatment. The clinical signs of FIC include variable combinations of dysuria, pollakiuria, hematuria, and periuria (inappropriate urination). FIC may be the most common cause of lower urinary tract signs (LUTS). It affects two thirds of the 1.5 per cent of cats that present to primary care veterinarians with LUTS.¹

This review suggests that FIC is not a single disease entity, presents the evidence for the comorbidity of a variety of unexplained clinical conditions in some patients with FIC, describes recent experimental results obtained in cats with severe FIC that reveal the presence of an underlying neuroendocrine abnormality in these patients, and shows how these results offer the possibility of additional therapeutic approaches for this subset of patients.

One of the most valuable resources for understanding such a cryptic disorder is the availability of a naturally occurring disease model in another species that permits investigations not possible in the target species. The most relevant naturally occurring model of FIC identified to date is interstitial cystitis (IC), a chronic pelvic pain syndrome of human beings characterized by pain referable to the urinary bladder and urinary frequency and urgency. Two presentations of IC are recognized based on cystoscopic evaluation of the bladder (although the necessity of cystoscopy for the diagnosis of IC is debated among urologists).² As in cats, only submucosal petechial hemorrhages are observed in most IC patients (type I), whereas ulcers occur within the dome and lateral walls of the bladder in a minority (<20 per cent) of human patients (type II), and only rarely in cats. Studies from human beings with IC and other chronic pain syndromes have helped us gain additional insight into abnormalities we have documented in cats with FIC.

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Human patients with IC appear to suffer from a wide variety of comorbid conditions and the data suggest the same is true in cats. For example, LUTS have been reported to be a comorbid condition in cats with separation anxiety syndrome,³ hypertrophic cardiomyopathy,⁴ and obesity.⁵ These findings suggest that FIC may not be a disease process localized to the bladder. Further research in cats with FIC supports this hypothesis. Clinicians must obtain a thorough history and physical examination of cats with LUTS to identify other abnormalities and to help prepare specific therapeutic protocols for individual cats with FIC.

One important challenge to any hypothesis for a non-bladder etiology for FIC is, “Why are clinical signs related to the urinary bladder?” A more illuminating question may be, “Why can so many organ systems be affected in patients with FIC and IC?” The answer to this question may be related to alterations in the two primary stress response systems in the body: the sympathetic nervous system (SNS) and the hypothalamic-pituitary-adrenal (HPA) axis.

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Based on our research in cats with severe, recurrent FIC, we have documented an enhanced activation of the stress response system, primarily the SNS limb. A schematic diagram of the normal regulation of this complex neuroendocrine system⁶ is presented in Figure 47-1. In summary, once stimulated by higher brain structures responding to the perception of a threat, corticotrophin-releasing factor (CRF) is released from the hypothalamus, which acts as a hormone to stimulate the anterior pituitary gland to release ACTH. CRF also acts as a neurotransmitter to stimulate neurons in the brainstem, including the locus coeruleus, to activate the sympathetic nervous system. Under mild stress conditions, cats with FIC had significantly higher plasma levels of dihydroxyphenylalanine (DOPA), norepinephrine (NE), and other catecholamine metabolites

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